

Data Types: General

All data is held as a binary number := **datum**. But consider 0100 1000 0110 1001. The datatype tells us what this number actually is.

Data types come in two flavors: **primitive** and **composite**.

A primitive data type is a datatype which is stored as a single object in memory.

- integer
- float
- boolean
- char

Languages usually operate very quickly on these primitive data types.

Composite data types are composed of two or more primitive data types.

- date
- array (dynamic or static length)

Primitive Data Types

Integers

Integers are whole numbers, included negative integers and 0. Many languages distinguish between signed and unsigned integers, or have different integer types with different maximum values.

For example, in Java:

- byte (1 byte): -128 to 127
- short (2 bytes): -32768 to 32767
- int (4 bytes): -2147483648 to 2147483647
- long (8 bytes): -9223372036854775808 to 9223372036854775807

Real

Real or **floating point** data types cover numbers with a fractional part, like 6.7 or 3.14.

Many languages allow real data types with different maximum values for real numbers, and also with different precisions.

In python:

- float, with 53-bit precision, or about 15 decimal digits.

In Java:

- float (4 bytes), about 6/7 decimal digits.
- double (8 bytes), about 15 decimal digits.

Boolean

- Non-numeric
- Python uses a whole byte instead of a bit

String

Holds a sequence of encoded characters.

In some languages, strings are composite types ([char]) but in python, the string data type is primitive. Encoded using UTF-8.

Pointer

A **pointer** data type holds the address of a value in memory. So the value of the pointer is just the memory location, not the actual value.

Pointers are used in most C-like languages (e.g. C++, rust) to make it easier to navigate complex data structures.

However, **pointers do not exist in python.**

Strong and Weak Typing

In Python, variables can be declared as a datatype (e.g. `i: int = 2`), but you can change the datatype of the variable to a string without errors.

This is known as **weak typing**.

In other languages (like rust, C++), **strong typing** is used.

Nothing?

Some data types allow empty representations e.g. `[]` or `""`. Though these objects are ‘empty’, they still have the type `list` and `str` respectively.

The only datatype that is truly ‘nothing’ is `None` in python (of type `NoneType`).

Composite Data Types

Array

An **array** is an ordered group of elements of the same data type.

Like a `Vec` in Rust.

Record

A record is an ordered group of elements. These elements can be of different data types.

In Python, lists, dicts, and tuples are all records.

User-defined data types

A user-defined data type is a data type created by a user based on existing primitive and composite data types.

In python:

```
pet_dict = {'id': None, 'name': None, 'species': None, 'owner': None}
my_pet = pet_dict.copy()
my_pet['id'] = 23301
my_pet['name'] = 'Aggie'
my_pet['species'] = 'cat'
my_pet['owner'] = 'Mr Clark'
```

Subroutines

A **procedure** is a subroutine that does not return a value. A **function** is a subroutine that returns a one or more values.

Object-Oriented Programming

Inheritance

A subclass **inherits** the attributes and methods of its superclass by default. e.g. `class Foo extends Bar` or in python, for `class Foo(Bar)`, Foo inherits Bar.

Attributes and Methods

Unless it says otherwise, attributes and methods of a class are private by default.

Complex Data

Stack

A stack is a data structure which operates by “last in, first out”. Used in ideas such as “stack frame” -
> recursive functions use these.